

# Solutions

NCERT Class XII Chemistry · Chapter 1 · Solutions

## Solutions — Complete Notes

*15 marks on the blueprint — the heaviest chapter in the paper. Theory you know; the marks live in the numericals. Every formula below is followed by the method that actually gets used in the exam.*

### Types of solution — *nine combinations, one table*

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A solution is a homogeneous mixture of two or more components. Composition and properties are uniform throughout. Two components only = **binary solution**.

The component in the largest quantity is the **solvent** — and the solvent fixes the physical state of the solution. Everything else is **solute**.

| Solvent | Solute                 | Example                                                 |
|---------|------------------------|---------------------------------------------------------|
| Gas     | Gas                    | Air (O <sub>2</sub> in N <sub>2</sub> )                 |
| Liquid  | Chloroform in nitrogen |                                                         |
| Solid   | Camphor in nitrogen    |                                                         |
| Liquid  | Gas                    | O <sub>2</sub> in water; CO <sub>2</sub> in soft drinks |
| Liquid  | Ethanol in water       |                                                         |
| Solid   | Glucose in water       |                                                         |
| Solid   | Gas                    | H <sub>2</sub> in palladium                             |
| Liquid  | Amalgam (Na in Hg)     |                                                         |
| Solid   | Alloys — Cu in Au      |                                                         |

**Examiner asks:** one-markers naming the solute/solvent type for a given example. Camphor-in-nitrogen and hydrogen-in-palladium are the two people forget.

### Concentration terms — *seven of them — and which one the question wants*

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The whole game is reading which term the question gives you and which it wants. Get that right and the arithmetic is trivial.

#### Mass percentage (w/w)

$$\text{mass \%} = \frac{\text{mass of component}}{\text{mass of solution}} \times 100$$

"10% glucose by mass" means **10 g glucose in 100 g solution** — so 90 g water. Used for industrial chemicals.

### Volume percentage (v/v)

$$\text{volume \%} = \frac{\text{volume of component}}{\text{volume of solution}} \times 100$$

Used when both components are liquids. 35% v/v ethylene glycol is car antifreeze — it drops water's freezing point to 255.4 K.

### Mass by volume (w/v)

Mass of solute in 100 mL of solution. Medicine and pharmacy. **0.9% w/v NaCl = normal saline** — remember this one, it reappears in the osmosis questions.

### Parts per million

$$\text{ppm} = \frac{\text{parts of component}}{\text{total parts}} \times 10^6$$

For trace quantities — pollutants, dissolved oxygen in seawater.

### Mole fraction ( $x$ )

$$x_A = \frac{n_A}{n_A + n_B} \quad x_A + x_B = 1$$

No units. This is the one Raoult's and Henry's laws are written in.

### Molarity (M)

$$M = \frac{\text{moles of solute}}{\text{volume of solution in L}}$$

**Depends on temperature** — volume expands on heating, so molarity falls.

### Molality (m)

$$m = \frac{\text{moles of solute}}{\text{mass of solvent in kg}}$$

**Independent of temperature** — mass doesn't change with T. This is why every colligative-property formula uses molality, not molarity.

**Trap:** molarity uses *volume of solution*; molality uses *mass of solvent*. Not solution. Losing the solute mass from the denominator is the single most common arithmetic slip in this chapter.

Worked · 2014 — molality and molarity from mass %

10% by mass glucose ( $M = 180 \text{ g/mol}$ ) in water, density 1.2 g/mL. Find molality and molarity.

**Set up:** 10% by mass  $\rightarrow$  10 g glucose in 100 g solution  $\rightarrow$  water = 90 g.

**Molality:**

$$m = \frac{10/180}{90/1000} = 0.62 \text{ mol kg}^{-1}$$

**Molarity:** need volume. From  $d = m/V$ :  $V = 100/1.2 = 83.3 \text{ mL} = 0.0833 \text{ L}$ .

$$M = \frac{10/180}{0.0833} = 0.67 \text{ mol L}^{-1}$$

**Examiner asks:** "given mass % and density, find molality and molarity" is a standing 3-marker. The density is only ever there to get you the volume.

## **Solubility — *what dissolves in what, and what temperature does to it***

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**Solubility** = maximum amount of solute that dissolves in a specified amount of solvent at a specified temperature.

### **Solid in liquid**

**Like dissolves like.** Polar solutes (NaCl, sugar) dissolve in polar solvents (water); non-polar solutes (naphthalene, anthracene) dissolve in non-polar solvents (benzene).

Add solute and two opposing processes run: **dissolution** and **crystallisation**. When their rates become equal, dynamic equilibrium is reached, concentration goes constant, and the solution is **saturated**.

- **Temperature:** if dissolution is endothermic ( $\Delta H > 0$ ), solubility rises with T. If exothermic ( $\Delta H < 0$ ), solubility falls with T. (Le Chatelier.)
- **Pressure:** no significant effect — solids and liquids are incompressible.

### **Gas in liquid**

Solubility **increases with pressure**: more gas particles per unit volume above the solution → more strike the surface → more dissolve. Solubility **decreases with temperature** (dissolution of a gas is exothermic).

**Examiner asks:** "aquatic species are more comfortable in cold water" (2019) — because oxygen's solubility falls as water warms. Straight 1-2 marker.

## **Henry's law — *the gas-solubility law and its three applications***

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At constant temperature, the solubility of a gas in a liquid is directly proportional to the partial pressure of that gas above the liquid. Equivalently:

$$p = K_H x$$

where  $p$  is partial pressure of the gas,  $x$  its mole fraction in solution, and  $K_H$  the Henry's law constant.

**Read  $K_H$  backwards:** higher  $K_H$  means *lower* solubility.  $K_H$  rises with temperature — which is exactly why gases get less soluble in warm water.

Worked · 2020, repeated 2026 — solubility from Henry's law

Solubility of  $\text{CO}_2$  in water at 298 K under 760 mm Hg.  $K_H = 1.25 \times 10^6$  mm Hg.

Solubility here means mole fraction. From  $p = K_H x$ :

$$x = \frac{p}{K_H} = \frac{760}{1.25 \times 10^6} = 6.08 \times 10^{-4}$$

Mole fraction has no unit.

### The three applications

- **Carbonated drinks** — bottles sealed under high pressure to force  $\text{CO}_2$  into solution.
- **Deep-sea diving / bends** — high pressure underwater dissolves  $\text{N}_2$  into blood; on ascent it comes out as bubbles that block capillaries. Divers' tanks use helium-diluted air (11.7% He, 56.2%  $\text{N}_2$ , 32.1%  $\text{O}_2$ ) because helium's high  $K_H$  means it barely dissolves.
- **Anoxia at altitude** — low partial pressure of  $\text{O}_2 \rightarrow$  low oxygen in blood and tissue  $\rightarrow$  climbers feel weak and can't think clearly.

**Examiner asks:** "state Henry's law + calculate mole fraction" as one 2-3 marker, or the bends/anoxia reasoning as a standalone. Both are near-certain appearances.

## Raoult's law and vapour pressure — *volatile pairs, then non-volatile solutes*

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### Two volatile liquids

Partial vapour pressure of each component is proportional to its mole fraction *in the liquid*:

$$p_1 = p_1^\circ x_1 \quad p_2 = p_2^\circ x_2$$

$$p_{\text{total}} = p_1^\circ x_1 + p_2^\circ x_2$$

For the mole fraction in the **vapour phase**, use pressures instead of moles:

$$y_1 = \frac{p_1}{p_{\text{total}}}$$

Worked · 2023 — total vapour pressure of an ideal solution

$p_X^\circ = 120$  mm Hg,  $p_Y^\circ = 160$  mm Hg, equal moles of X and Y mixed, ideal solution. Find  $p_{\text{total}}$ .

Equal moles  $\rightarrow x_X = x_Y = 0.5$ .

$$p_{\text{total}} = (120)(0.5) + (160)(0.5) = 60 + 80 = 140 \text{ mm Hg}$$

## Non-volatile solute in a volatile solvent

Only the solvent contributes to vapour pressure, so  $p_{\text{solution}} = p_1^\circ x_1$ . Adding a non-volatile solute always *lowers* vapour pressure — solute particles occupy surface area that solvent molecules would otherwise escape from.

## Raoult's law as a special case of Henry's law

Henry's law is  $p = K_H x$ ; Raoult's is  $p = p^\circ x$ . When the solute-solvent interaction happens to equal the solute-solute interaction,  $K_H$  becomes equal to  $p^\circ$  and the two laws coincide.

## Ideal and non-ideal solutions — *two deviations, with the examples that prove them*

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### Ideal solution

Obeys Raoult's law over the entire concentration range, and:

$$\Delta H_{\text{mix}} = 0 \quad \Delta V_{\text{mix}} = 0$$

Happens when A-A, B-B and A-B interactions are all nearly equal. Examples: n-hexane + n-heptane, bromoethane + chloroethane, benzene + toluene.

### Positive deviation

A-B interaction **weaker** than A-A and B-B. Molecules escape more easily → vapour pressure *higher* than Raoult predicts.

$$p_A > p_A^\circ x_A, \quad \Delta H_{\text{mix}} > 0, \quad \Delta V_{\text{mix}} > 0$$

Examples: ethanol + acetone (ethanol's own H-bonding is broken up), carbon disulphide + acetone.

### Negative deviation

A-B interaction **stronger** than A-A and B-B. Molecules held tighter → vapour pressure *lower* than predicted.

$$p_A < p_A^\circ x_A, \quad \Delta H_{\text{mix}} < 0, \quad \Delta V_{\text{mix}} < 0$$

Examples: phenol + aniline, chloroform + acetone (they form an H-bond between them), nitric acid + water.

**Trap:** chloroform + acetone is negative deviation because a *new* hydrogen bond forms *between* the two. Ethanol + acetone is positive because an *existing* hydrogen bond is broken. Both are asked; the reasoning direction is what earns the mark.

## Azeotropes

Binary mixtures with the **same composition in liquid and vapour phase**, boiling at constant temperature — so they cannot be separated by fractional distillation.

- **Minimum boiling azeotrope** ← large *positive* deviation. Ethanol + water, 95% ethanol by volume.
- **Maximum boiling azeotrope** ← large *negative* deviation. Nitric acid + water, 68% HNO<sub>3</sub> by mass, boils at 393.5 K.

**Examiner asks:** "define azeotrope + which type from negative deviation, with example" — a 2-3 marker that has run repeatedly.

## Colligative properties — *four of them, four formulas, one idea*

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Properties that depend on the **number** of solute particles, not their identity. That one sentence is the reason every question in this section works.

### 1 • Relative lowering of vapour pressure

$$\frac{p_1^\circ - p_1}{p_1^\circ} = x_2 = \frac{n_2}{n_1 + n_2}$$

For a dilute solution  $n_1 \gg n_2$ , so this simplifies to the working form:

$$\frac{p_1^\circ - p_1}{p_1^\circ} = \frac{w_2/M_2}{w_1/M_1}$$

### 2 • Elevation of boiling point

$$\Delta T_b = K_b m = \frac{K_b w_2 \times 1000}{M_2 w_1}$$

$K_b$  = molal elevation constant (ebullioscopic constant), units K kg mol<sup>-1</sup>. For water,  $K_b = 0.52$ .

### 3 • Depression of freezing point

$$\Delta T_f = K_f m = \frac{K_f w_2 \times 1000}{M_2 w_1}$$

$K_f$  = molal depression constant (cryoscopic constant). For water,  $K_f = 1.86$ , freezing point 273 K.

Both constants depend only on the solvent, and can themselves be calculated:

$$K_f = \frac{R M_1 T_f^2}{1000 \Delta H_{\text{fus}}} \quad K_b = \frac{R M_1 T_b^2}{1000 \Delta H_{\text{vap}}}$$

### 4 • Osmotic pressure

**Osmosis:** solvent flows through a semi-permeable membrane from pure solvent into solution.  
**Osmotic pressure** ( $\pi$ ) is the extra pressure applied on the solution side that just stops that flow.

$$\pi = CRT = \frac{n_2 RT}{V} = \frac{w_2 RT}{M_2 V}$$

**Why it's preferred for macromolecules** (proteins, polymers): works at room temperature, so heat-unstable biomolecules survive; uses molarity, which is easy to measure; and its magnitude is large even for dilute solutions, which is all you get from a poorly soluble polymer.

**Trap:**  $\pi$  uses *molarity* and volume in litres.  $\Delta T_b$  and  $\Delta T_f$  use *molality* and solvent mass in kg. Mixing these up is the standard way to lose an otherwise-correct 3-marker.

### Isotonic, hypertonic, hypotonic

- **Isotonic** — same osmotic pressure, no net osmosis. Blood cells are isotonic with 0.9% w/v NaCl.
- **Hypertonic** — higher concentration outside; water leaves the cell; cell **shrinks**.
- **Hypotonic** — lower concentration outside; water enters; cell **swells** and may burst.

**Reverse osmosis:** apply pressure *greater than*  $\pi$  on the solution side and solvent flows backwards, out of the solution. Used to desalinate seawater, with a cellulose acetate membrane that passes water but not ions.

## Abnormal molar mass and van't Hoff factor — *when the measured mass is wrong, and by how much*

Colligative properties give a molar mass. When the solute associates or dissociates, that measured value differs from the true one — an **abnormal molar mass**.

### Association

Molecules combine → fewer particles → colligative property falls → measured molar mass comes out **too high**. Ethanoic acid in benzene dimerises through hydrogen bonds: measured 120 g/mol against a true 60.

### Dissociation

Electrolyte splits into ions → more particles → colligative property rises → measured molar mass comes out **too low**. KCl gives 37.25 g/mol against a true 74.5.

### van't Hoff factor

$$i = \frac{\text{normal molar mass}}{\text{abnormal molar mass}} = \frac{\text{observed colligative property}}{\text{calculated colligative property}}$$

Equivalently, particles after dissociation/association ÷ particles before.

- **Dissociation** →  $i > 1$ . KCl: 2.  $\text{MgSO}_4$ : 2.  $\text{K}_2\text{SO}_4$ : 3.  $\text{CaCl}_2$ : 3.
- **Association** →  $i < 1$ . Complete dimerisation gives  $i = 0.5$ .
- **Neither** →  $i = 1$ . Glucose, urea, sucrose.

Every colligative formula then carries  $i$ :

$$\frac{p_1^\circ - p_1}{p_1^\circ} = i x_2 \quad \Delta T_b = i K_b m \quad \Delta T_f = i K_f m \quad \pi = \frac{i n_2 R T}{V}$$

## Partial dissociation or association

$$\text{dissociation: } i = 1 + (n - 1)\alpha \quad \text{association: } i = 1 + \left(\frac{1}{n} - 1\right)\beta$$

$\alpha$  = degree of dissociation,  $\beta$  = degree of association,  $n$  = particles produced or combined.

**Trap:** a degree of dissociation cannot exceed 1. If your  $\alpha$  comes out at 1.056, the intended answer is "essentially complete dissociation" — say so rather than writing an impossible number.

## Numerical patterns, collected — *every calculation type in this chapter, one model each*

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Six patterns cover essentially every numerical this chapter sets. Method first, then one worked model.

A · Concentration conversion — 3 marks

*Recognise it: gives a mass %, asks for molality/molarity, and hands you a density you'd otherwise have no use for.*

1. Take 100 g of solution. Solute mass = the percentage; solvent mass = 100 – that.
2. Moles of solute =  $w_2/M_2$ .
3. Molality: divide by solvent mass in kg.
4. Molarity: get volume from  $V = \text{mass}/d$ , convert to litres, divide.

B · Henry's law — 2 marks

*Recognise it: a  $K_H$  value appears.*

1.  $x = p/K_H$ . Answer is a mole fraction, unitless. That's the whole question.

C · Total vapour pressure — 3 marks

*Recognise it: two  $p^\circ$  values for two volatile liquids.*

1. Find mole fractions in the liquid (equal moles  $\rightarrow$  0.5 each).
2.  $p_{\text{total}} = p_1^\circ x_1 + p_2^\circ x_2$ .
3. If it asks for vapour-phase composition:  $y_1 = p_1/p_{\text{total}}$ .

D ·  $\Delta T_b / \Delta T_f \rightarrow$  molar mass or temperature — 3 marks

*Recognise it: a  $K_b$  or  $K_f$  is given. This is the most-set numerical in the chapter.*

1. Decide whether  $i$  is needed — is the solute an electrolyte, or does it dimerise?
2.  $\Delta T = iK \frac{w_2 \times 1000}{M_2 w_1}$ . Rearrange for whatever's missing.
3. If asked for the actual boiling/freezing point, *add*  $\Delta T_b$  to  $T_b^\circ$  or *subtract*  $\Delta T_f$  from  $T_f^\circ$ .  
Don't stop at  $\Delta T$ .

Worked · 2018 — freezing point of a glucose solution

60 g glucose ( $M = 180$ ) in 250 g water,  $K_f = 1.86$ . Find the freezing point.

Glucose doesn't dissociate, so  $i = 1$ .

$$\Delta T_f = \frac{1.86 \times 60 \times 1000}{180 \times 250} = 2.48 \text{ K}$$

$$T_f = 273 - 2.48 = 270.52 \text{ K}$$

E · Osmotic pressure — 3 marks

*Recognise it:  $R$  is given, or the word isotonic appears.*

1.  $\pi = \frac{w_2 RT}{M_2 V}$ , volume in **litres**.
2. For isotonic pairs, set the two concentrations equal:  $\frac{w_1}{M_1} = \frac{w_2}{M_2}$  (volumes cancel) and solve for the unknown  $M$ .

F · van't Hoff factor  $\rightarrow \alpha$  or  $\beta$  — 3-5 marks

*Recognise it: asks for degree of dissociation/association, or "predict the nature of the solute".*

1. Get  $i$  from the colligative-property equation with  $i$  left in as the unknown.
2.  $i > 1 \rightarrow$  dissociation, use  $i = 1 + (n - 1)\alpha$ .  $i < 1 \rightarrow$  association, use the association form.
3. Read  $n$  off the actual dissociation equation — write it out, don't guess.

Worked · 2026 — degree of dissociation of fluoroacetic acid

19.5 g  $\text{FCH}_2\text{COOH}$  ( $M = 78$ ) in 500 g water gives  $\Delta T_f = 1 \text{ K}$ .  $K_f = 1.86$ .

$$1 = i \times \frac{1.86 \times 19.5 \times 1000}{78 \times 500} \Rightarrow i = 1.075$$

$\text{FCH}_2\text{COOH} \rightarrow \text{FCH}_2\text{COO}^- + \text{H}^+$ , so  $n = 2$ :

$$1.075 = 1 + (2 - 1)\alpha \Rightarrow \alpha = 0.075 = 7.5\%$$

## Past year questions · question types — ranked by how often they turn up

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*From the last 10 years of boards. 22 questions are captured below with their years; the source video states it covers 25, so treat this as the bulk of the set rather than all of it.*

1 · Colligative-property numerical — 3 marks

*Recognise it:  $K_b$ ,  $K_f$  or  $R$  is supplied.*

1. Pick the right formula from the constant given.
2. Decide on  $i$ .
3. Convert to the final temperature if asked.

**Trap:** stopping at  $\Delta T$  when the question asked for the boiling or freezing point.

2 · Deviation from Raoult's law, with reason — 2-3 marks

*Recognise it: names a specific liquid pair and asks "what type of deviation, why".*

1. Compare A-B interaction against A-A and B-B.
2. Stronger → negative, vapour pressure falls. Weaker → positive, vapour pressure rises.
3. Name the actual interaction — which H-bond forms or breaks.

**Trap:** answering "negative deviation" without saying *which* bond forms between the two. The reason carries the mark.

3 · van't Hoff factor / degree of dissociation — 3-5 marks

*Recognise it: asks for  $i$ ,  $\alpha$ ,  $\beta$ , or "predict association or dissociation".*

1. Solve the colligative equation for  $i$ .
2. Compare against 1 to decide association vs dissociation.
3. Apply the matching  $\alpha/\beta$  relation.

**Trap:** using  $n = 2$  automatically.  $\text{CaCl}_2$  gives 3 ions,  $\text{K}_2\text{SO}_4$  gives 3.

4 · Henry's law statement + calculation — 2-3 marks

*Recognise it:  $K_H$  appears, often paired with "state Henry's law".*

1. State the law in mole-fraction form.
2.  $x = p/K_H$ .
3. If asked for applications, give carbonated drinks and the bends.

5 · Osmosis reasoning (cells, meat, isotonic) — 2-3 marks

*Recognise it: blood cells, salting meat, or a % NaCl solution.*

1. Compare concentration inside vs outside.
2. Water moves toward the higher concentration.
3. State the consequence: shrink, swell, or preserved.

6 · Definitions and two-difference questions — 2 marks

*Recognise it: "define...", "write two differences between..."*

1. Give the definition in the book's own words.
2. For differences, answer in pairs — ideal vs non-ideal on both  $\Delta H_{\text{mix}}$  and  $\Delta V_{\text{mix}}$ .

## Past year questions · mark slots — *what each type is worth, and the time that buys*

| Question type             | Slot                           | Time    |
|---------------------------|--------------------------------|---------|
| Colligative numerical     | 3-marker                       | 4–5 min |
| Deviation + reason        | 2–3 marker                     | 3 min   |
| van't Hoff / $\alpha$     | 3-marker, sometimes inside a 5 | 5 min   |
| Henry's law               | 2–3 marker                     | 3 min   |
| Osmosis reasoning         | 2–3 marker                     | 3 min   |
| Definitions / differences | 2-marker                       | 2 min   |
| Concentration conversion  | 3-marker                       | 4 min   |

*Slots marked from the question shapes in the paper — the video groups them as short-answer type I (2 marks) and type II (3 marks).*

## Past year questions · repeat offenders — *appeared more than once — highest probability in the chapter*

These are the reason the PYQ video is worth watching. Each has run at least twice.

2020 Q3 · 2026 Q1

Henry's law statement plus mole fraction of  $\text{CO}_2$  at 760 mm Hg — the identical numbers reappeared six years apart.

2019 Q11(a) · lecture PYQ

Aquatic species more comfortable in cold water — gas solubility falls as temperature rises.

2019 Q11(b) · 2019 Q12

Anoxia at high altitude, and chloroform + acetone negative deviation. Both recur as reason-type one/two markers.

2018 Q10 · 2026 Q18

Freezing-point depression with a given  $K_f$  — glucose in water, then fluoroacetic acid in water with a degree of dissociation on top.

2023 Q5 · lecture PYQ

Total vapour pressure of an ideal solution from two  $p^\circ$  values and equal moles. Same 120/160 mm Hg numbers in both.

2016 Q7 · 2025 Q6

van't Hoff factor: definition, and why osmotic pressure is preferred for macromolecular molar masses.

2017 Q13 · 2020 Q2

Blood cells in 1.2% and 0.4% NaCl, and why 0.1 M KCl boils higher than 0.1 M glucose. Both test "colligative = number of particles".

## Past year questions · cold practice — *answers only* — *work them before you look*

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One model per pattern is above. These are the same patterns with different numbers: work each one cold, then check.

### Colligative numericals

- 2026 Q14 — 8 g non-volatile solute in 100 g diethyl ether, bp rises  $35.60 \rightarrow 36.86$  °C,  $K_b = 2.02$ . Find  $M_2$ . 128.25 g/mol
- 2017 — 10 g  $\text{CaCl}_2$  ( $M = 111$ ) in 200 g water,  $K_b = 0.52$ . Find  $\Delta T_b$ . 0.702 K (remember  $i = 3$ )
- 2026 Q8 — 1 molal trichloroacetic acid boils at 100.18 °C,  $K_b = 0.512$ . Find  $i$ . 0.35
- 2021 Q15 — 2.56 g sulphur in 100 g  $\text{CS}_2$ ,  $\Delta T_f = 1.62$  K,  $K_f = 4.9$ ,  $M(\text{S}_8) = 256$ . Find % association.  $i = 0.846$ ; 15.4%

### Vapour pressure

- 2026 Q19 — 61 g benzoic acid in 500 g benzene,  $p^\circ = 51.2$  mm Hg, complete dimerisation. Find  $p_s$ . 50.42 mm Hg (use  $i = 0.5$ )
- Lecture PYQ — pure water  $p^\circ = 32$  mm Hg at 308 K falls to 31.84 with 10 g solute in 200 g water. Find  $M_2$ . 180 g/mol

### Osmotic pressure

- 2020 Q3 — 5% urea solution ( $M = 60$ ) at 300 K,  $R = 0.0821$ . Find  $\pi$ . 20.51 atm
- 2019 Q17 — 6% glucose ( $M = 180$ ) isotonic with 2.5% of an unknown. Find the unknown's  $M$ . 75 g/mol

### Freezing point, two-stage

- 2021 Q21 — 5% cane sugar ( $M = 342$ ) freezes at 271 K. Find the freezing point of 5% glucose ( $M = 180$ ). 269.2 K — find  $K_f$  from the sugar first

Built from Sourabh Raina's Solutions one-shot and PYQ videos, cross-checked against NCERT Class XII Chemistry Chapter 1 (Rationalised 2022-23). Constants verified: water  $K_b = 0.52$ ,  $K_f = 1.86$  K kg mol<sup>-1</sup>.